



HILBERT'S 10th PROBLEM IN LOW DIMENSIONS

J. Maurice Rojas*
Texas A&M University
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HILBERT'S $10^{\text{th}} + \varepsilon$

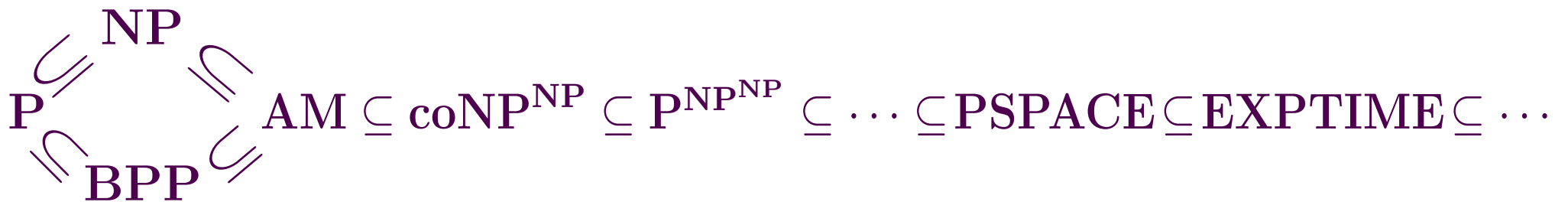
- $\text{FEAS}_R :=$ “Decide whether an input system of polynomials in $\mathbb{Z}[x_1, \dots, x_n]$ has a root in R^n .”

HILBERT'S $10^{\text{th}} + \varepsilon$

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FEAS_R

$\sqsupseteq$
?

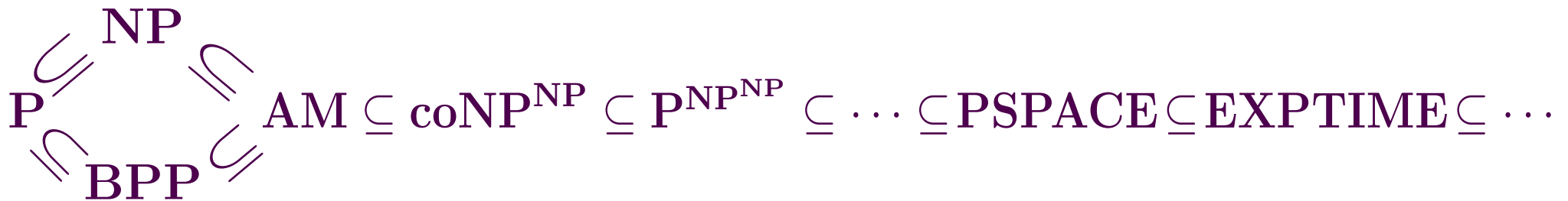


REVIEW

- **FEAS_ℤ** := “Decide whether an input system of polynomials in $\mathbb{Z}[x_1, \dots, x_n]$ has a root in $\boxed{\mathbb{Z}^n}$.”

FEAS_ℤ

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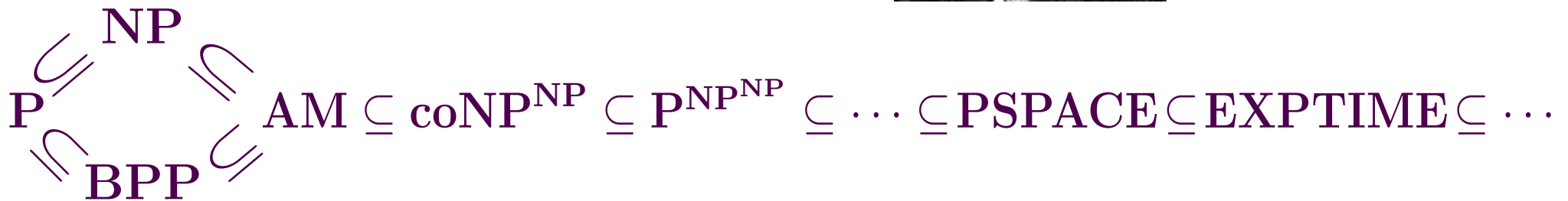


REVIEW

- **FEAS_ℤ** := “Decide whether an input system of polynomials in $\mathbb{Z}[x_1, \dots, x_n]$ has a root in $\mathbb{Z}^n$.”

FEAS_ℤ

⊞
?



THE HIGH END

- **FEAS $\mathbb{Z}$** := “Decide whether an input system of polynomials in $\mathbb{Z}[x_1, \dots, x_n]$ has a root in $\boxed{\mathbb{Z}^n}$.”



Julia Robinson in 1950

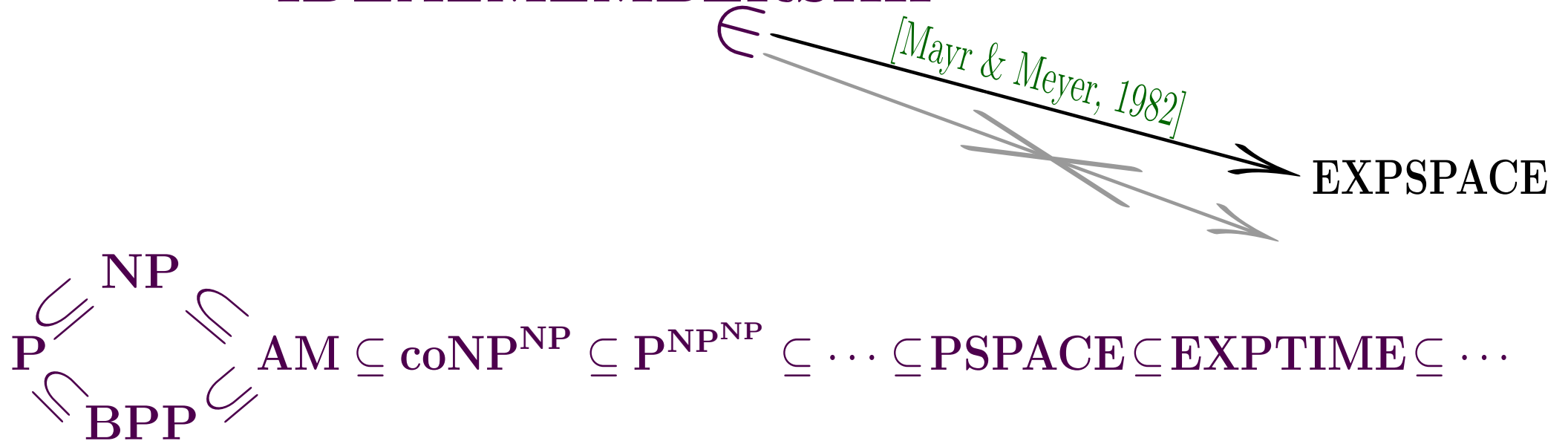


FEAS $\mathbb{Z}$ $\in$ $\xrightarrow[\text{[Matiyasevic, 1970]}]{\text{[Davis, Putnam, Robinson, 1961]}}$ **RE**

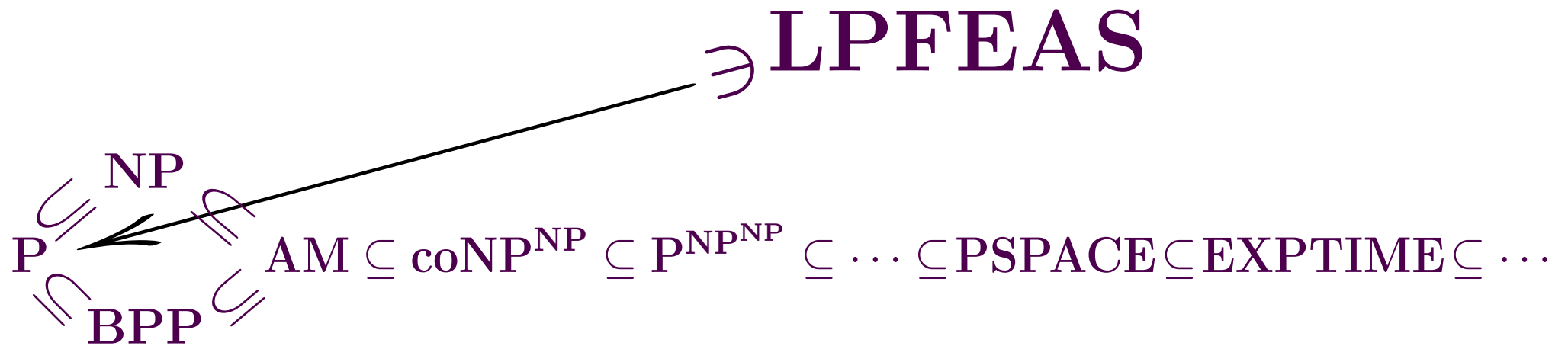
$\begin{matrix} & \text{NP} \\ \text{P} & \subseteq & \\ & \text{BPP} \end{matrix}$
 $\text{AM} \subseteq \text{coNP}^{\text{NP}} \subseteq \text{P}^{\text{NP}^{\text{NP}}} \subseteq \dots \subseteq \text{PSPACE} \subseteq \text{EXPTIME} \subseteq \dots$

THE HIGH END

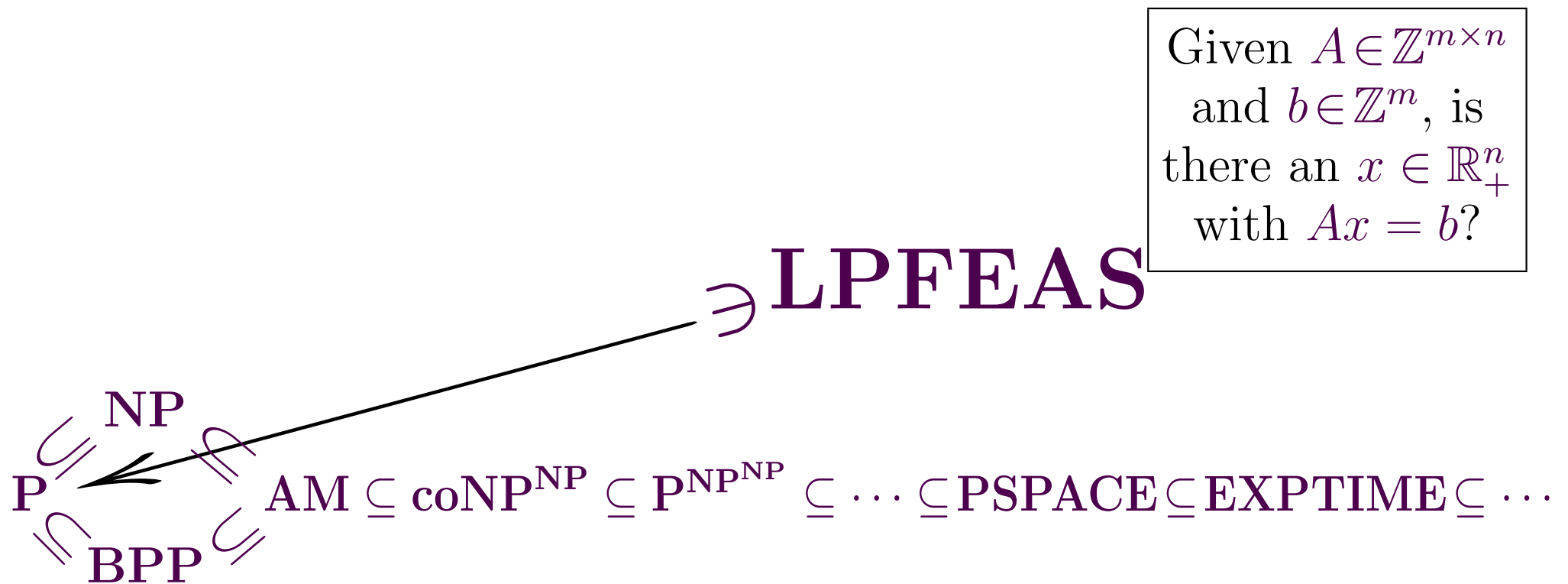
IDEALMEMBERSHIP



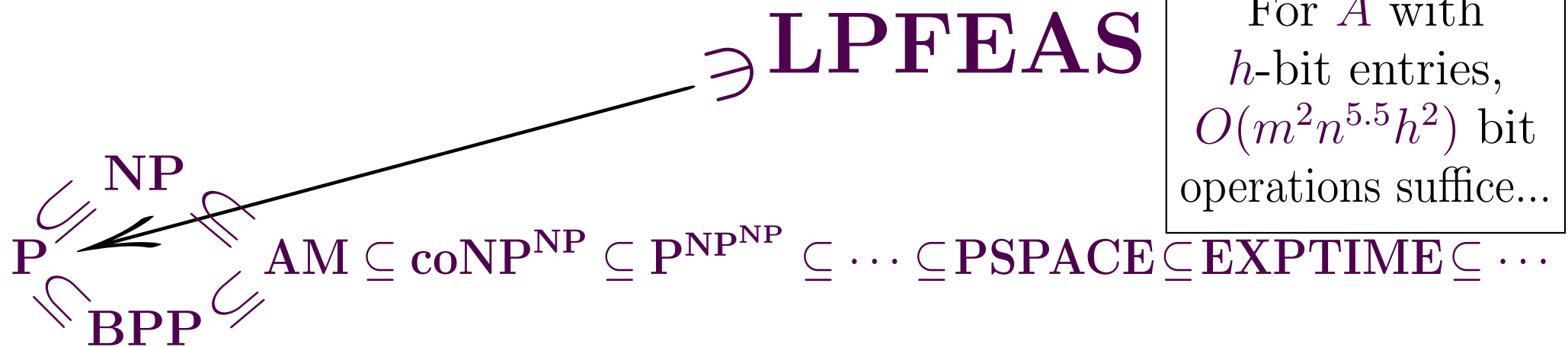
THE LOW END



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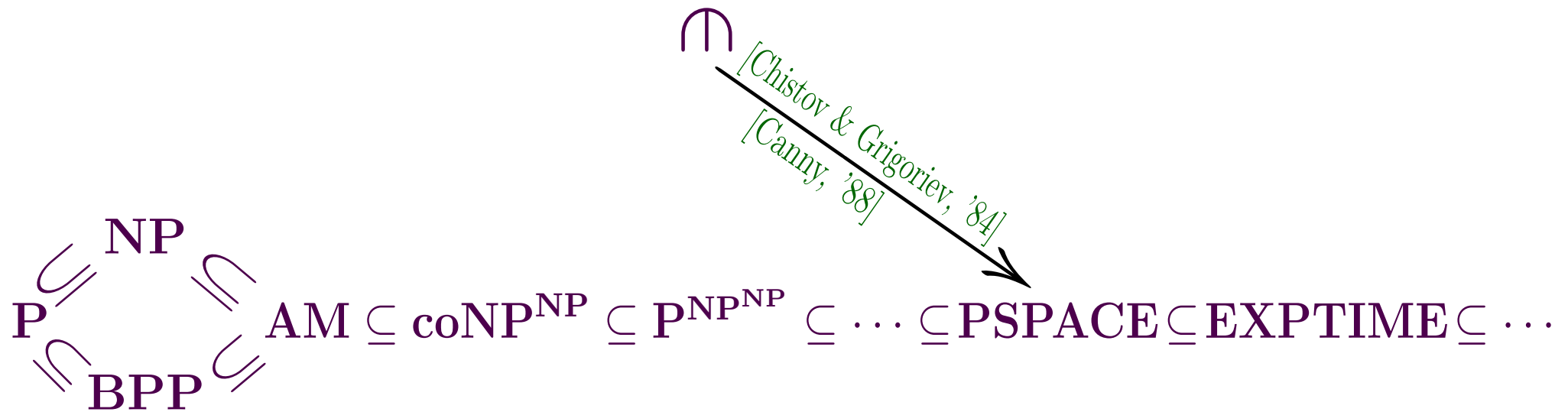
Given $A \in \mathbb{Z}^{m \times n}$
and $b \in \mathbb{Z}^m$, is
there an $x \in \mathbb{R}_+^n$
with $Ax = b$?

[Karmarkar, '84]:
For A with
 h -bit entries,
 $O(m^2 n^{5.5} h^2)$ bit
operations suffice...

$\mathbb{C}$ and $\mathbb{R}$

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$\text{FEAS}_{\mathbb{C}}, \text{FEAS}_{\mathbb{R}}$



BACKGROUND: NP...

Recall that **3CNFSAT** is the problem of deciding whether a **particular** class of logical expressions has a *satisfying assignment*, e.g., is

$(y_2 \vee \neg y_3 \vee y_1) \wedge (\neg y_2 \vee \neg y_3 \vee y_5) \wedge (y_2 \vee y_3 \vee \neg y_5) \wedge (\neg y_2 \vee y_3 \vee \neg y_4) \wedge (y_1 \vee \neg y_2 \vee \neg y_4)$
satisfiable?

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satisfiable?

For this example, yes: (T, F, T, F, F) works...

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NP-hard := “at least as hard” as any problem in **NP**.

NP-complete := **NP-hard** and in **NP**.

NP VIA 3CNFSAT

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[Cook-Levin, 1971-1972]

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[Cook-Levin, 1971-1972]

[Karp, 1972] then found reductions between dozens of computational problems, later inspiring a **huge** zoo of **NP**-complete problems...

NP VIA FINITE FIELDS

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$\supseteq$ Knottedness! (via $\mathfrak{M} = S^3$ and $g = 0$)

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